INCH-POUND

MIL-C-24640/13C
26 June 1995
SUPERSEDING
MIL-C-24640/13B
31 July 1989

MILITARY SPECIFICATION SHEET

CABLE, ELECTRIC, 600 VOLTS, LIGHT WEIGHT, MULTI PAIR AWG 26, TYPE 2XSX0

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-C-24640.

CONSTRUCTION
(Non-Watertight)

FIRST - 19 strand copper conductor in accordance with ASTM B-286, tinned strands in accordance with ASTM B-33. AWG size 26 maximum diameter 0.022 inches.

SECOND - Component wires shall be in accordance with MIL-W-81044/12 and tested in accordance with the component wire ratings and requirements listed on this slash sheet. Quality conformance testing shall be in accordance with Table VI of MIL-C-24640.

THIRD - Two conductors, one black and one white, twisted together. The pitch shall be 3/4 ± 1/8 inch.

FOURTH - Binder tape applied helically with overlap, at manufacturer's option.

FIFTH - Braided shield of # 38 AWG tin coated copper strands. Braid coverage shall be no less than 90% determined in accordance with MIL-C-27500.

SIXTH - Shield insulation of two polyester tapes or polyester tape and extruded jacket. Identification codes applied using Method 2.

SEVENTH - Four shielded pairs shall be cabled together. Cabling sequence shall be consecutive. Fillers may be used as necessary, to form a round cable.

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EIGHTH - Binder tape applied helically with overlap.

NINTH - Braided shield, tin coated copper strands. Braid must be constructed to meet EMP requirements as listed herein.

TENTH - Binder tape applied helically with overlap, at manufacturer's option.

ELEVENTH - Crosslinked polyolefin jacket. (See Table I for thickness).

TABLE I. Construction details.

PIN M24640/13	Type number	Number of pairs	Diameter over shield (inch) (nom)	Jacket thickness (inch) (minimum	Cable diameter (inch) min	Cable diameter (inch) max	Cable weight (pounds per 1000 feet) (max)
-01UO	2XSX0-4	4	0.243	0.038	0.333	0.359	101.0

REQUIREMENTS:

Qualification required.
Construction details: See tables I.

Component wire:

For the component wire the following additional tests shall be applicable in accordance with MIL-W-81044:

WIRE, RATINGS AND ADDITIONAL REQUIREMENTS:

Temperature rating: 150°C (302°F) maximum conductor temperature
Voltage rating: 600 volts (rms) at sea level.
Accelerated aging: Oven temperature, 300± 2°C (572± 3.6°F) for 6 hours.
Blocking: 150± 2°C (302± 3.6°F)
Flammability: 30 seconds (max); 3.0 inches (max); no flaming of tissue paper.
Humidity resistance: 5000 megohms for 1000 feet, minimum insulation resistance after humidity exposure.
Impulse dielectric test:
Primary insulation (when tested in lieu of spark test): 6.0 kilovolts (peak) 100 percent test.
Finished wire: 8.0 kilovolts (peak), or 3 Khz spark test 5.5KV, 100 percent test.

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WIRE, RATINGS AND ADDITIONAL REQUIREMENTS (continued):

Insulation resistance: 5000 megohms for 1000 feet (min).
Life cycle: Oven temperature, 200± 2°C (392± 3.6°F) for 168 hours.
Physical properties of primary insulation:
Tensile strength, 2500 lb/in^2 (min).
Elongation, 150 percent (min).
Shrinkage: 0.125 inch maximum at 300± 2°C (572± 3.6°F).
Spark test primary insulation: 1500 volts (rms), 60 Hz, 100 percent test.
Surface resistance: 500 megohms inches (min), initial and final readings.
Thermal shock: Oven temperature, 150± 2°C (302± 3.6°F).
Maximum change in measurement: 0.060 inch.
Wet dielectric test: 2500 volts (rms).
Wrap test: Mandrel wrap test required.

EXAMINATION AND TESTS:

	Basic electrical:	Requirements
	Conductor Resistance - ohms/1000 feet at 25°C, maximum	44.24
	Voltage withstand - volts, root mean square, minimum	
*	Conductor to conductor and shield	2000
	Conductor to shield	Not applicable
*	Shield to shield	200
*	Component shield to overall shield	500
	Insulation resistance - megohms/1000 feet, minimum	
*	Conductor to shield	2500
*	Shield to shield (component shield only)	100
*	Shield to shield (component shield to overall shield)	100
	Conductor and shield continuity	No failure
	Jacket flaws	No failure
	Group A:	
	Visual and dimensional	No failure
	Capacitance:	
	Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum	Not applicable
	Capacitance unbalance - percent, maximum	Not applicable
	Characteristic impedance - at 1 MHz, ohms	Not applicable
	Watertightness (see table XI of MIL-C-24640 for limits of water leakage)	Not applicable
	Weight	No failure
	Group B:	
	Cold bending	No failure
	Drip	Not applicable
*	Cross link proof test (percent maximum)	
	Jacket	50
	Tear - pounds per inch thickness, minimum	35

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	Group B (continued):	
	Physicals (unaged)	
	Jacket	
	Tensile strength - lb/in^2, minimum	1300
	Elongation - percent, minimum	160
	Attenuation - at 3 MHz, dB/100 feet, maximum	Not applicable
	Gas flame (1 hour)	Not applicable
	Group C:	
	Physicals (aged)	
	Jacket	
	Air oven	
	Tensile strength - percent of unaged, minimum	60
	Elongation - percent of unaged, minimum	60
	Hot oil immersion	
*	Tensile strength - percent of unaged, minimum	50
*	Elongation - percent of unaged minimum	50
*	Permanence of printing (conductor) - cycles, minimum	Not applicable
*	Permanence of printing (jacket) - cycles, minimum	125
	Armor - conformance to material, construction and coverage	Not applicable
	Shield - conformance to material, construction and coverage	No failure
	Heat distortion - percent of unaged, maximum	30
*	Cable sealant removability	Not applicable
	Shrinkage	No failure
	Mutual inductance - dB/1000 feet, maximum	Not applicable
*	Surface transfer impedance	
	Milliohms per meter, maximum	700
	EMP response, dB, minimum	60
	Pulse response time - ms/500 feet	
	Rise time, maximum	Not applicable
	Decay time, maximum	Not applicable
*	Group D	
	Flame propagation	No Failure
	QUALIFICATION INSPECTION:	
	Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:	
	Aging and compatibility (cable)	No failure
	Acid gas equivalent - percent, maximum	
	Jacket	2
	Fillers	2
	Insulation	18
	Halogen content - percent, maximum	
	Jacket	0.2
	Fillers	0.2

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QUALIFICATION INSPECTION (continued):

	Immersion	
	Tensile strength - percent of unaged, minimum	50
	Elongation - percent of unaged minimum	50
*	Smoke index, maximum	
	Jacket	25
	Fillers	45
	Insulation	10
*	Toxicity index, maximum	
	Jacket	5
	Fillers	5
	Insulation	1.5
	Durometer (jacket) - (type A) hardness, minimum	80
	Weathering (jacket)	No failure
	Fungus Resistance	No failure
	Abrasion resistance	No failure
	Gas Flame (3 hour)	Not applicable
	Component wire	No failure

Unit ordering length: 500 feet nominal.

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Navy - SH

Preparing Activity:
Navy - SH

Review Activities:
Army - AV, CR, ER, ME, AR, AL
Navy - EC, CG
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Agent:
DLA-IS

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